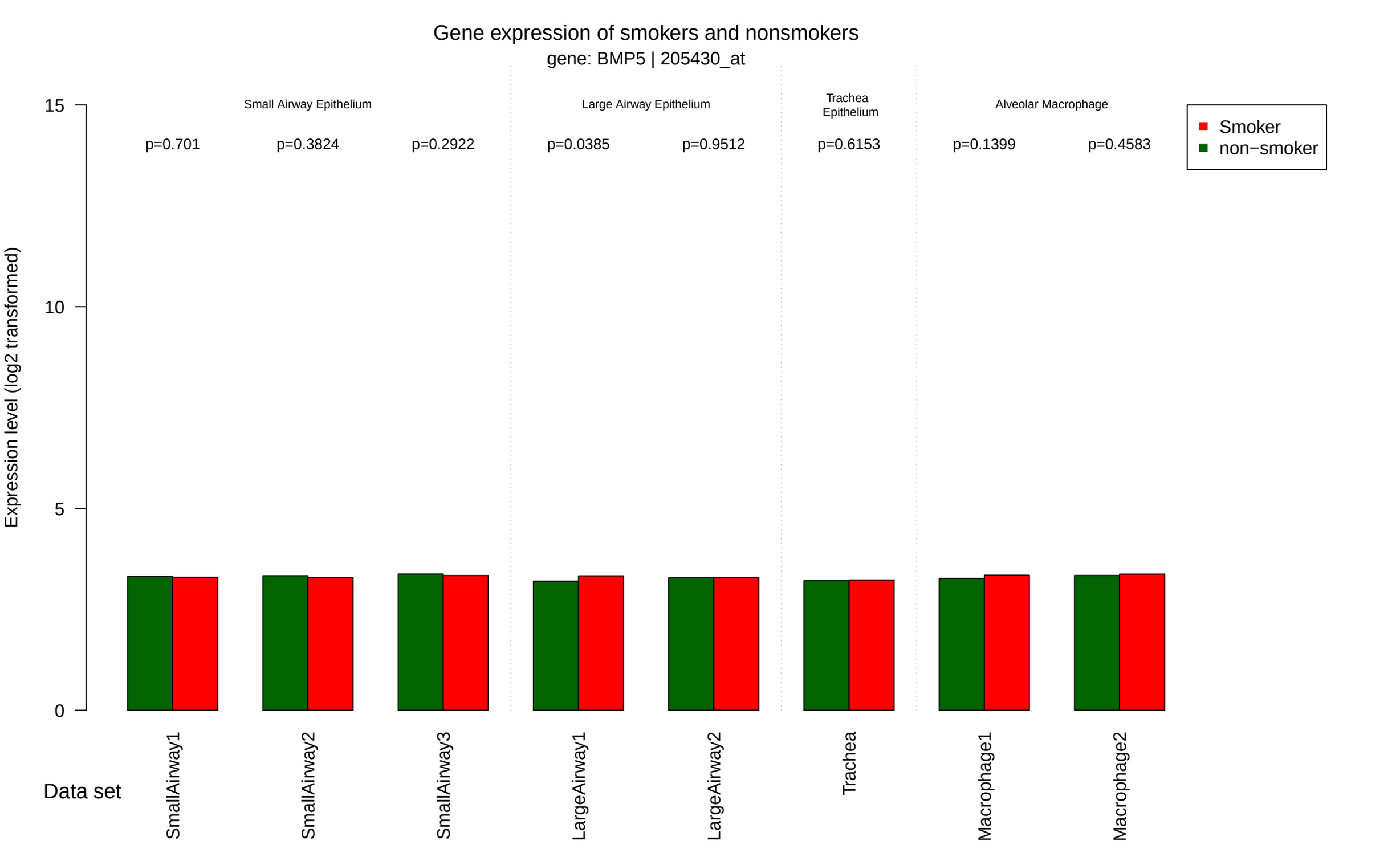
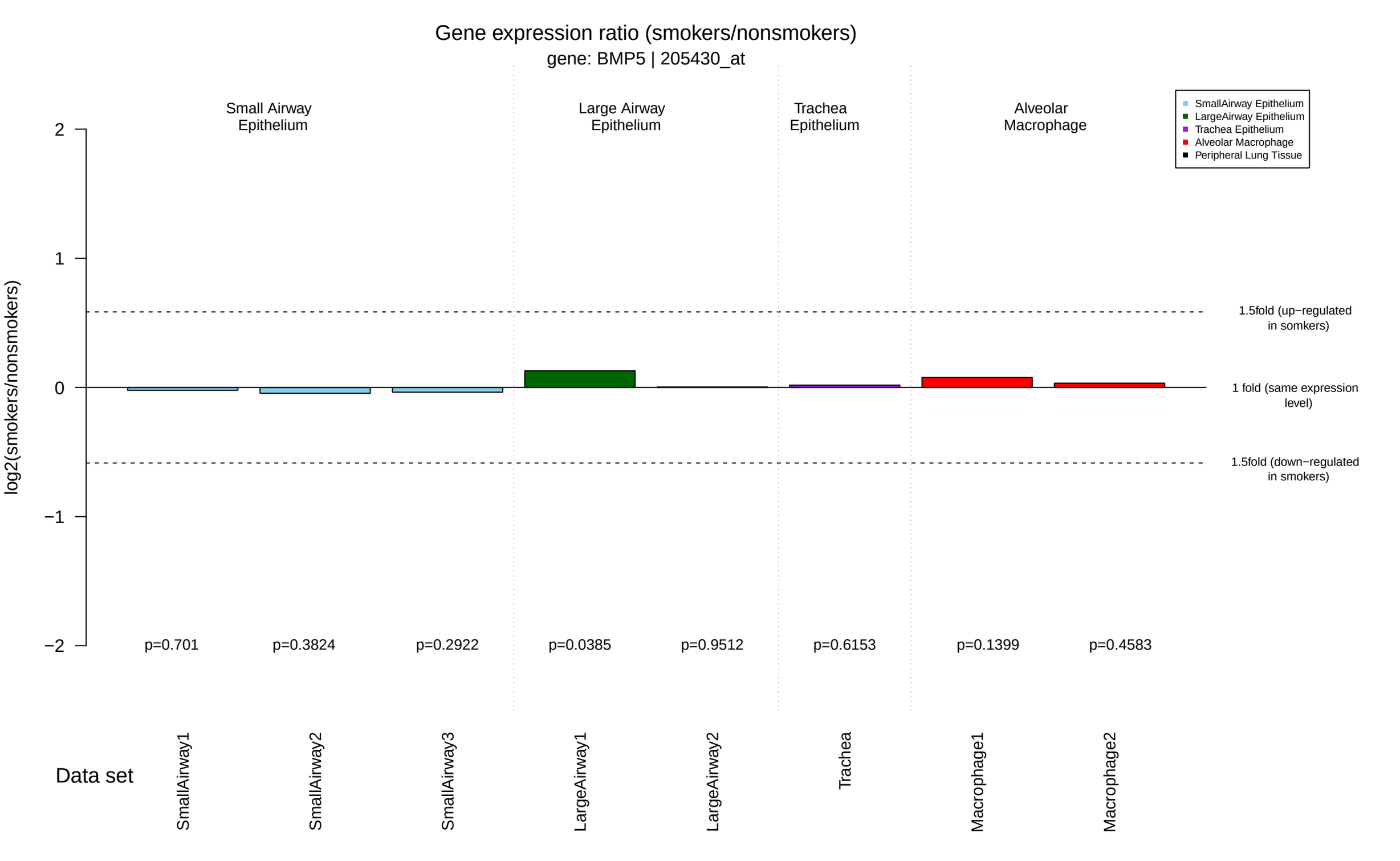
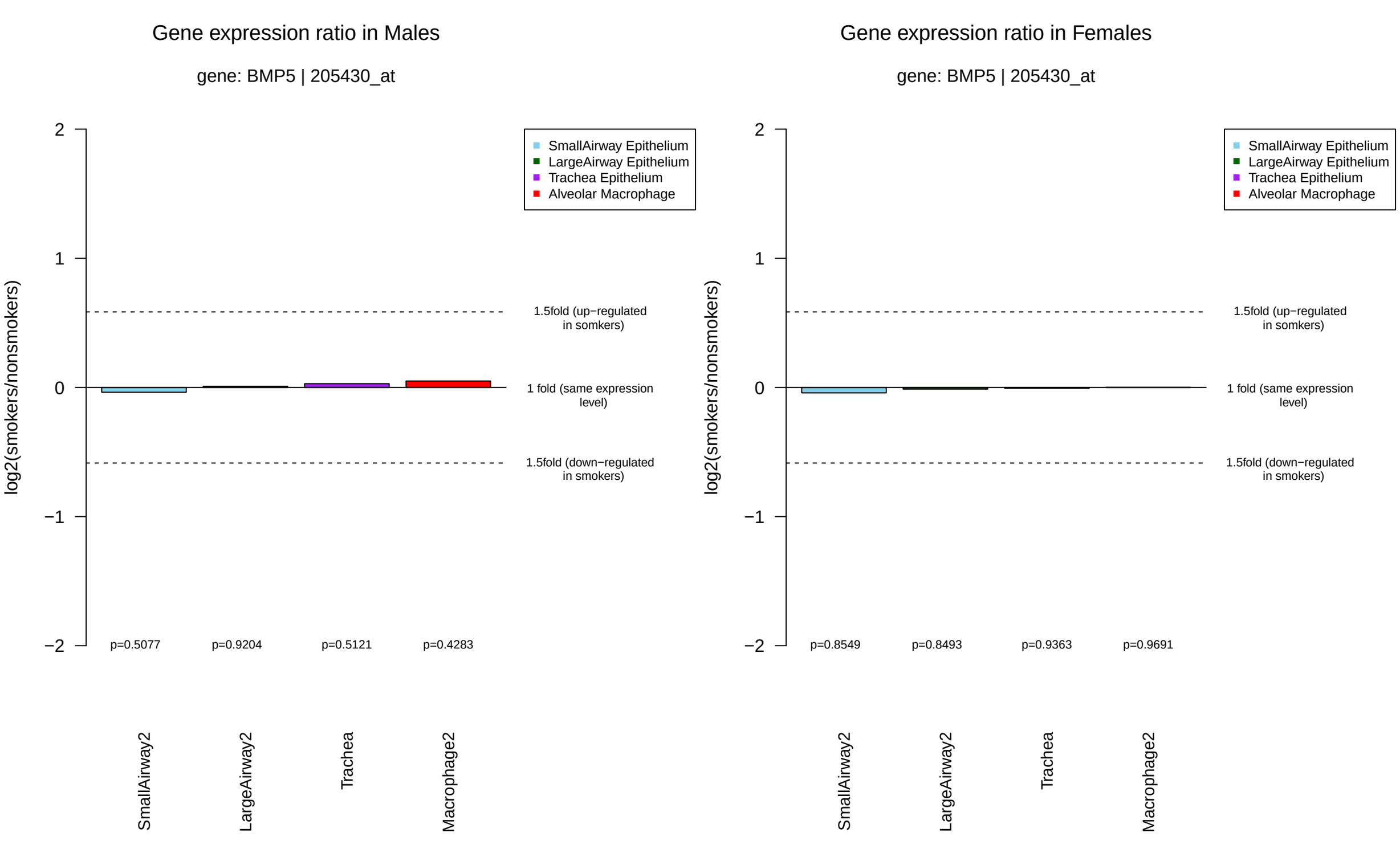
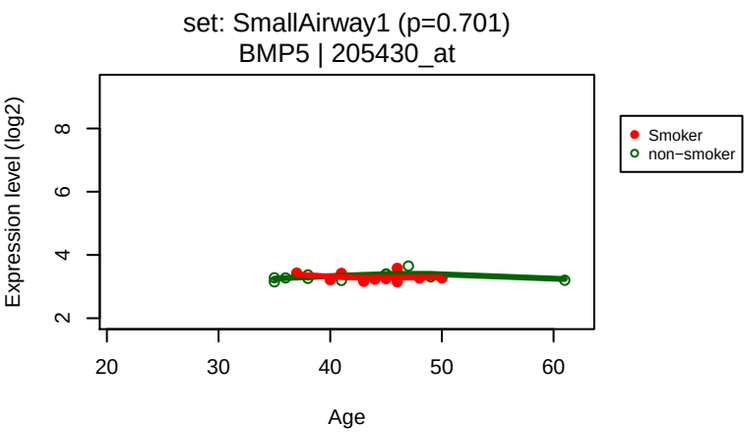
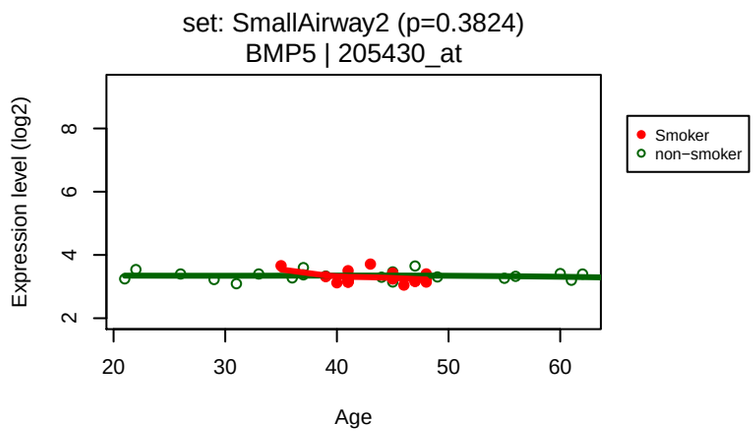
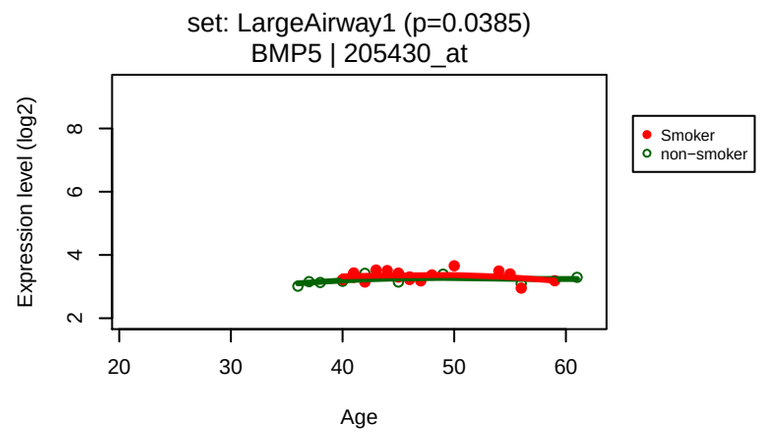

Gene expression of smokers and nonsmokers
gene: BMP5 | 205430_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.701
p=0.3824
p=0.2922
p=0.0385
p=0.9512
p=0.6153
p=0.1399
p=0.4583
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

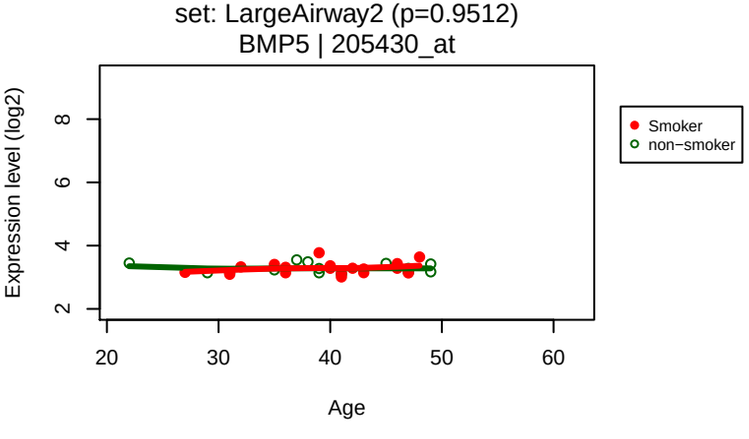

Gene expression ratio (smokers/nonsmokers)
gene: BMP5 | 205430_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.701
p=0.3824
p=0.2922
p=0.0385
p=0.9512
p=0.6153
p=0.1399
p=0.4583
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

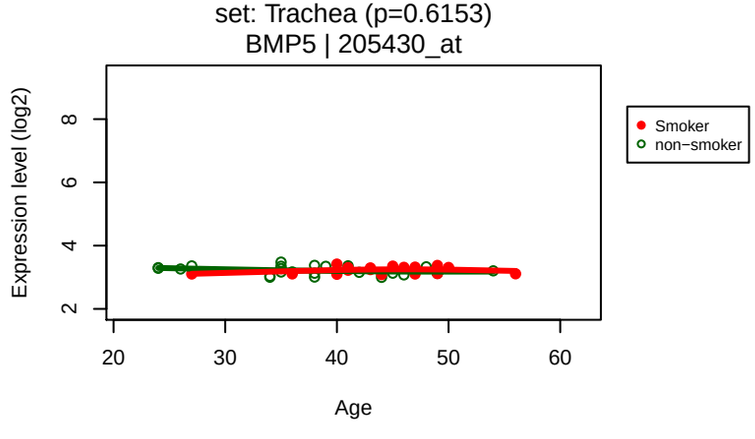

Gene expression ratio in Males
gene: BMP5 | 205430_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5077
p=0.9204
p=0.5121
p=0.4283
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BMP5 | 205430_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8549
p=0.8493
p=0.9363
p=0.9691
SmallAirway2
LargeAirway2
Trachea
Macrophage2

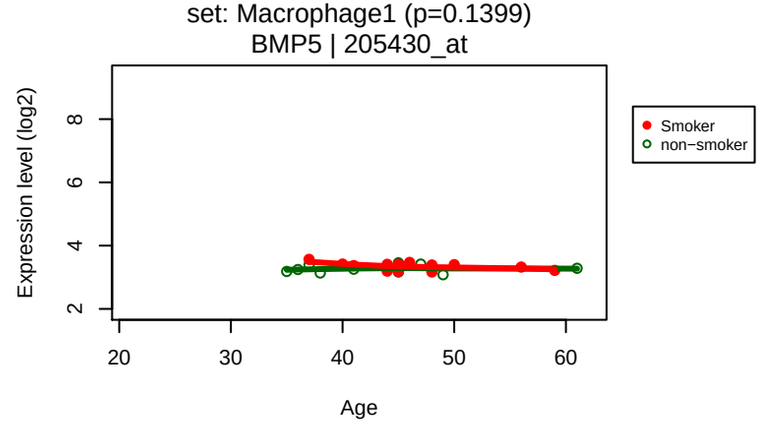

set: SmallAirway1 (p=0.701)
BMP5 | 205430_at
Expression level (log2)
Age
Smoker
non−smoker

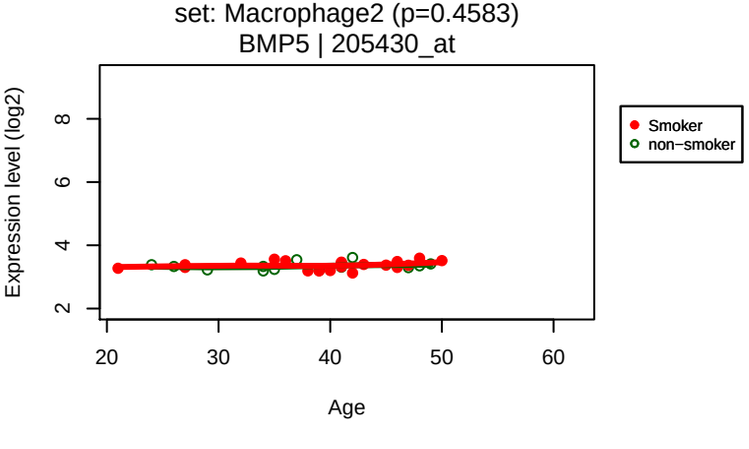

set: SmallAirway2 (p=0.3824)
BMP5 | 205430_at
Expression level (log2)
Age
Smoker
non−smoker

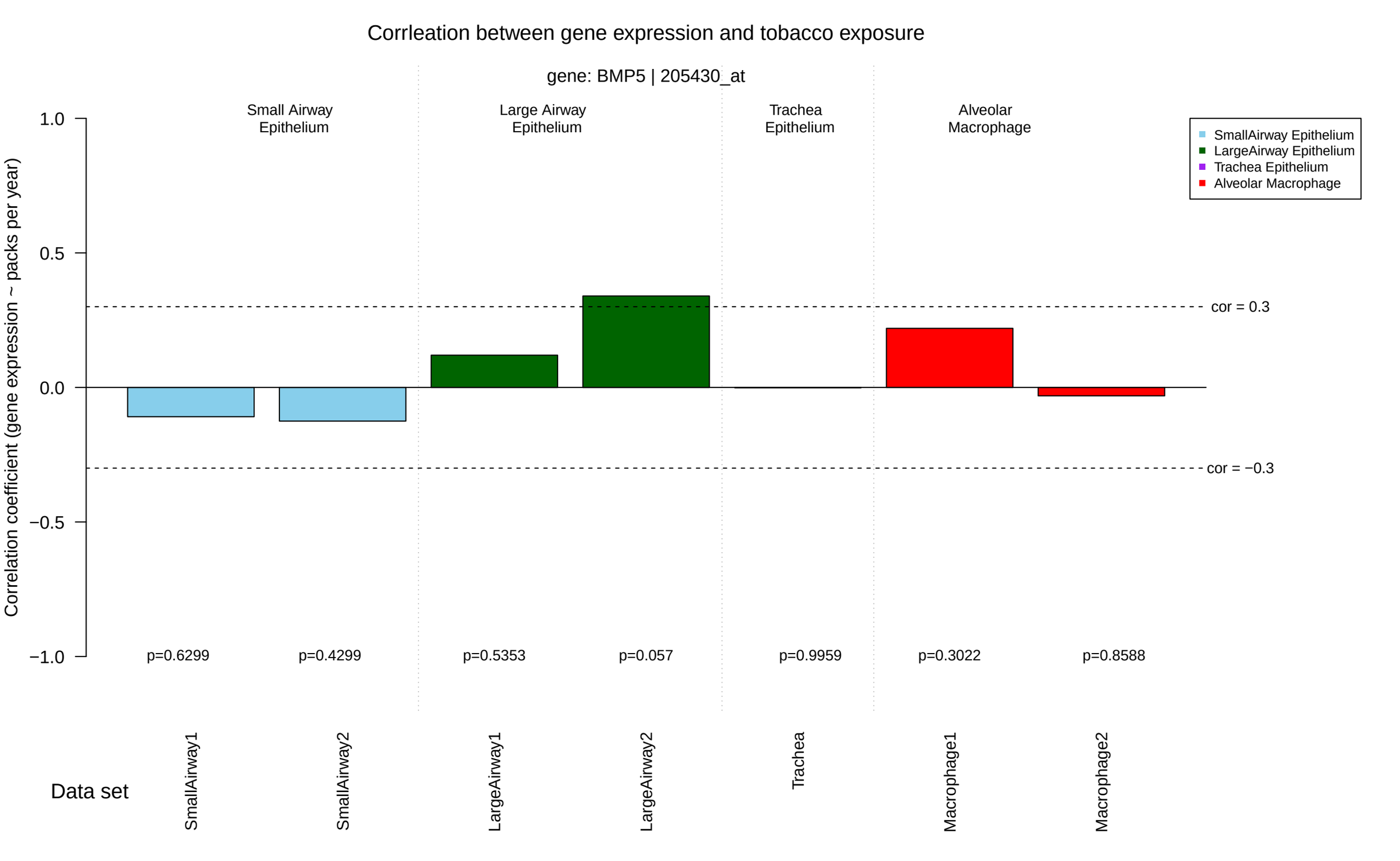

set: LargeAirway1 (p=0.0385)
BMP5 | 205430_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9512)
BMP5 | 205430_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6153)
BMP5 | 205430_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1399)
BMP5 | 205430_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4583)
BMP5 | 205430_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BMP5 | 205430_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6299
p=0.4299
p=0.5353
p=0.057
p=0.9959
p=0.3022
p=0.8588
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2